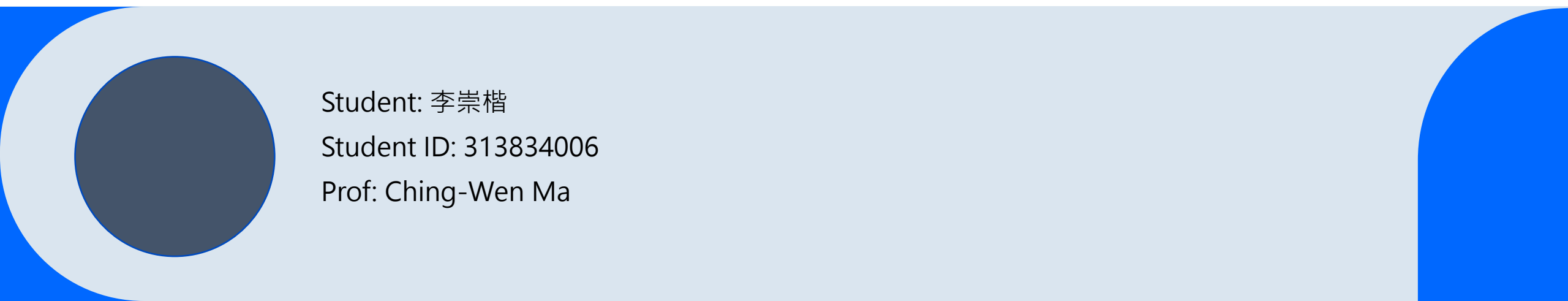




Final Project Proposal : Real-Time Email Monitoring and Alert System Based on a Self-Trained Spam Detection Model



Student: 李崇楷

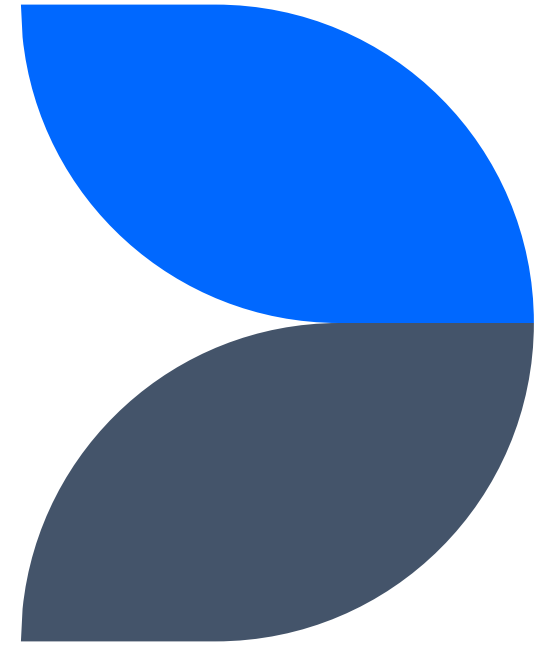
Student ID: 313834006

Prof: Ching-Wen Ma

Outline

- Introduction
- Structure
- Method
- Reference

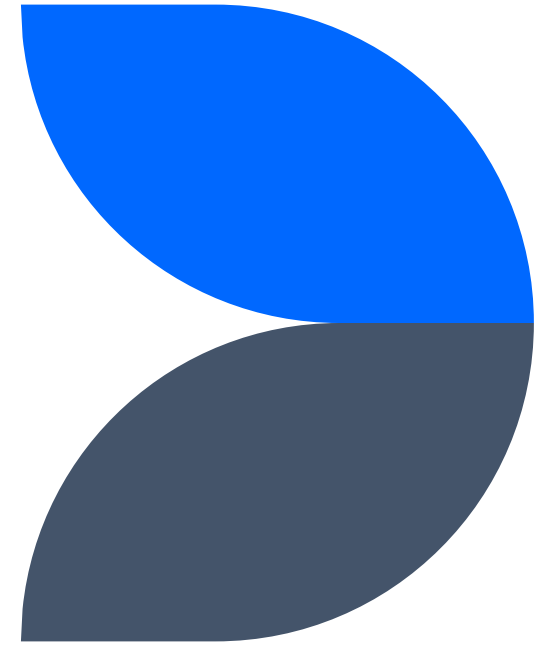
Introduction



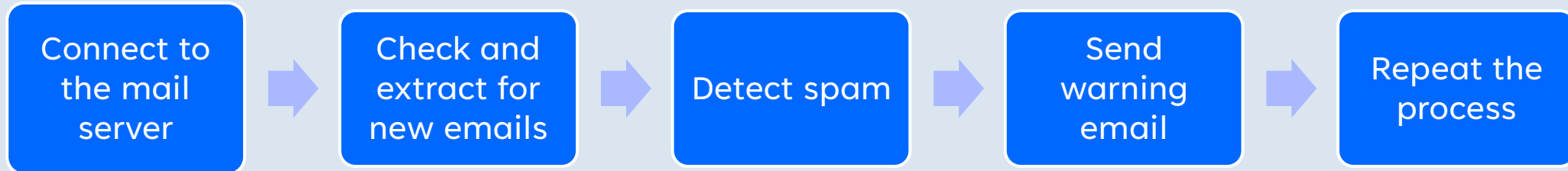
Introduction

While Gmail and other email providers offer spam detection, they are not foolproof and often fail to catch sophisticated phishing attempts. To address this, we developed an email monitoring system using a custom-trained spam detection model. By leveraging the IMAP protocol, our system fetches emails, analyzes them, and alerts users if a message is flagged as spam or phishing. This ensures enhanced protection by addressing the limitations of existing spam filters with a tailored machine-learning approach.

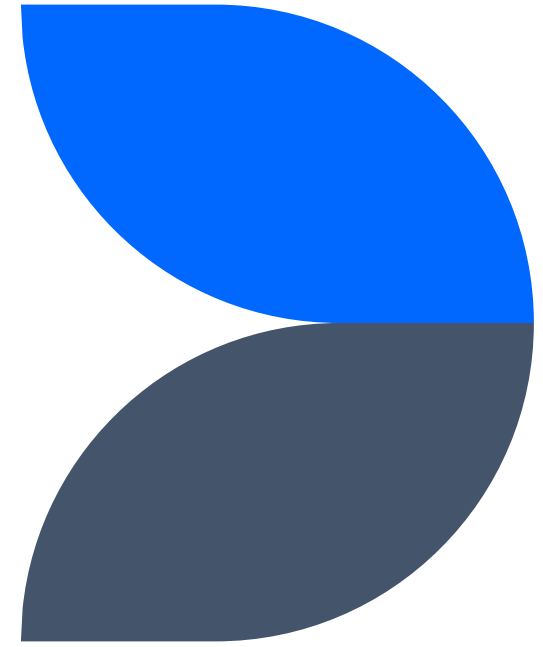
Structure



Structure



Method



Connect to the Mail Server

IMAP (Internet Message Access Protocol) is a protocol used for remotely accessing and managing emails from a mail server. It allows users to view folder structures, download email metadata or partial content (such as attachments), and perform actions (such as marking emails as read, moving, or deleting emails).

```
# Connect to Gmail's IMAP server
imap = imaplib.IMAP4_SSL("imap.gmail.com")
imap.login(username, password)
print("Login successful!")
```

```
Connecting to IMAP server...
Login successful!
```

Check and Extract for New Emails

```
# Select the mailbox folder
imap.select(folder)
print(f"Selected folder: {folder}")

# Search for all emails
status, messages = imap.search(None, "ALL")
mail_ids = messages[0].split()[-5:]
print(f"Found {len(mail_ids)} emails to process")
```

```
Found 5 emails to process
Fetching email 1/5...
Fetching email 2/5...
Fetching email 3/5...
Fetching email 4/5...
Fetching email 5/5...
Fetching emails complete.
```

Subject: 您已經繳交 Week11 (11/15) Seminar Reflection Submission 作業了

Body: 1131.639406 ->作業 ->Week11 (11/15) Seminar Reflection Submission您已經繳交了作業“Week11 (11/15) Seminar Reflection Submission” 點擊以查看您 繳交的作業。

Subject: 【陽明交通大學 校園公告系統】2024/11/15 新增訊息公告

Body: 今日新增之公告 近 3 天到期之公告 當日活動公告 今日新增之公告 到期日 主旨 公告單位 公告類別 2024/11/28 中山醫學大學附設醫院於113年12月7日至12月8日舉辦「2024大數據醫學研究國際研討會」。研究發展處企劃一組 電子

Subject: 【Amazon X NYCU】113年11月27日 16:30-18:00 Amazon X NYCU ECE X CTLD Tech Talk 2024

Body: Amazon ◆ NYCU ECE ◆ CTLD Tech Talk 2024 報名: <https://forms.gle/ygphhbWuJBwxeBKF9> 活動日期: 113年11月27日 活動時間: 16:30-18:00 活動地點: 交大校區工程四館116教室 ☀ 講師: Amazon團隊 ★★★★★講座大綱

Subject: 服務學習中心【國際志工成果靜態展】活動開跑啦~歡迎大家來觀展! Service-Learning Center 【International Volunteer Achievement Static Exhibition】is starting!

Body: 2024 國立陽明交通大學印度、東南亞國際志工團成果靜態展啟動囉!【我們去了哪些地方?】【今年志工團們做了些什麼?】快揪好朋友們一起來看展,一口氣帶你們了解~【活動資訊】展期:2024/11/1-2024/11/29地點:浩然圖書館2F

Subject: 安全性快訊

Body: 用於登入帳戶的應用程式密碼已產生 kevin611116.ai13@nycu.edu.tw 假如您並未替「IMAP」產生這個密碼,表示可能有人盜用了您的帳戶。請立即檢查帳戶並採取必要保護措施。查看活動您也可以前往以下網址查看帳戶安全相關活動

Detect spam by Model

- Dataset: [\[1\] Spam Email](#)

Category		Message
ham spam	87% 13%	5157 unique values
ham		Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got a...
ham		Ok lar... Joking wif u oni...

- Model: [\[2\] LSTM](#) [\[3\] Decision Tree](#) [\[4\] Logistic Regression](#)
- Reference: [\[5\] spam emails classification\(ACC 99%\)](#)
[\[6\] email_classification_96%accuracy](#)

Dataset Preprocess

Origin texts = ["Hello world!", "Spam messages are extremely annoying.", "Good morning, have a nice day."]
Origin labels = ["ham", "spam", "ham"]



Processed Texts: ['hello world', 'spam message extremely annoying', 'good morning nice day']



Vectorized Texts: [[0 0 0 0 1 0 0 0 0 1] [1 0 1 0 0 1 0 0 1 0] [0 1 0 1 0 0 1 1 0 0]]

Encoded Labels: [0 1 0]

Feature Names: ['annoying' 'day' 'extremely' 'good' 'hello' 'message' 'morning' 'nice' 'spam' 'world'] (ngram_range =(1,1))

Send warning email

SMTP (Simple Mail Transfer Protocol) is a communication protocol used for transmitting emails over the network. Its primary purposes are sending emails and forwarding emails, and it can support multiple formats like plain text, HTML, and emails with attachments.



您可能收到可疑的郵件或訊息，請務必提高警覺！詐騙郵件通常會：

- 要求您點擊可疑連結。
- 要求提供您的銀行帳號、密碼或其他個人資料。
- 包含讓您感到緊急的內容，例如：「帳戶即將被凍結」或「贏得大獎」。

請勿點擊任何不明連結，並仔細核對郵件發件人地址。如有疑慮，請聯繫官方客服。



保護您的安全是我們的責任，請時刻保持警惕。

Reference

[\[1\] Spam Email](#)

[\[2\] LSTM](#)

[\[3\] Decision Tree](#)

[\[4\] Logistic Regression](#)

[\[5\] spam emails classification\(ACC 99%\)](#)

[\[6\] email_classification_96%accuracy](#)

謝謝大家

